

Component	Median (ms)	Mean (ms)	Std Dev (ms)	IQR (ms)	5th pct (ms)	95th pct (ms)	Skewness	% of Total	Shapiro-W p
ECDH P-256	0.1369	0.1428	0.0189	0.0095	0.1313	0.1683	5.141	29.1%	< 0.001
ML-KEM-512	0.3258	0.3350	0.0326	0.0165	0.3136	0.3719	6.667	69.2%	< 0.001
KDF (SHA-256)	0.0047	0.0052	0.0024	0.0004	0.0044	0.0074	15.154	1.0%	< 0.001
Total Workflow	0.4708	0.4830	0.0466	0.0249	0.4519	0.5358	5.802	100%	< 0.001

Table V.4: Hybrid Workflow Component-Level Latency - 2,000 Iterations

AES Config	Enc Median (ms)	Dec Median (ms)	Throughput (MB/s)	Enc IQR (ms)	Enc Skewness
AES-256-CBC 1 KB	0.0170	0.0174	57.44	0.0009	14.036
AES-256-CBC 1 MB	3.2915	3.3014	303.81	0.3223	1.682

Table V.5: AES-256-CBC Performance Using Hybrid-Derived Session Key — 200 Iterations